

Section 2.1: Manual Input Devices

In this section we will look at how input devices can be used to get data into a computer system.

In section 2.1 we will focus on manual input devices and then move onto direct input devices in section 2.2.

For each input device you have to be able to **identify it from an image**, explain what it is **used for** and discuss **advantages/disadvantages**.

Learning objectives of this section:

- # Know the **purpose of** and be able to **identify** different manual input devices.
- # Understand the **uses of** each manual input device.
- # Be able to **discuss advantages and disadvantages** of each manual input device.

Input Devices (what are they?)

Definition: "Hardware devices that allow data to be entered into a computer."



Key Words:

Input, Data, Manual, Automatic

Overview of Input Devices:

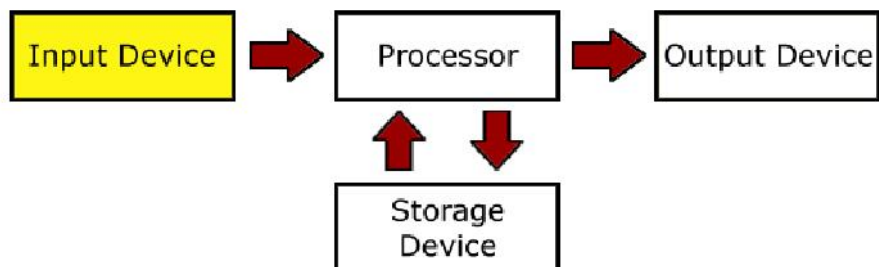
- # Input devices are part of the 4 main hardware components of a computer system:
 - **Input Devices** - these get data into the computer
 - **Processor** - this does something with the data to make it useful information
 - **Output Devices** - these show the results of processing
 - **Storage Devices** - holds the data in the system.
- # The image below shows where input devices fit into a computer system:

This diagram
shows where input

Examples:



A mouse is an example of a commonly used



devices fit into a computer system.

input device.



A keyboards is another example of a commonly used input device.

There are two different types of input devices:

1. **Manual Input Devices -** Data is input into the computer **by hand** (people have to do most of the work)
2. **Direct Input Devices -** Data is input into the computer directly by a **machine/device** (there is minimal input by people)

NOTE:

More examples will be shown further down this page.

Manual Input Devices (data entered by hand)



Key Words:

Keyboard, Keypad, Pointing Device, Remote Control, Joystick, Touch Screen, Scanner, Digital Camera, Microphone, Graphic Tablet, Video Camera, Web Cam, Light Pen

Examples of Manual Input Devices:

The table below shows you some examples of manual input devices. Click each device to move to more information:

Keyboard



Numeric Keypad



Pointing Devices



Remote Control



REMEMBER!

Manual input devices require humans to do most of the work needed to get data into the system.

Joystick**Touch Screen****Scanner****Graphics Tablet****For example:**

If you are using a keyboard to input instructions into the computer, you have press the keys yourself. It is a manual job not automatic.

Light Pen**Microphone****Digital Camera****Web Camera****NOTE:**

Click the images to the left to move to more information about each one.

Breakdown of each manual input device:

The following sections will help summarise each of the manual input devices. For each one you will see:

- **Information** about the device
- **Uses** of the device
- **Advantages** of the device
- **Disadvantages** of the device

1 Keyboard

Information:

Keyboards are one of the most commonly used manual input devices.

Key boards are used to input the following into the computer:

- **Text**
- **Numbers**
- **Instructions**

Most keyboards use a **QWERTY** key layout. The name '**QWERTY**' comes from the first five lettered keys on on the top row of the keyboard - Q-W-E-R-T-Y.

Examples:



The first five lettered keys of a keyboard spell out QWERTY.

- # Special keyboards called '**Ergonomic Keyboards**' have been developed to help **reduce health problems** (such as repetitive strain injury) linked with typing.

Ergonomic keyboards have a more **natural shape** to them with reduces stress on the wrist and hands.

Ergonomic keyboards also come with a **built-in hand-rest** which further helps prevent health issues such as **RSI** (Repetitive Strain Injury).



Ergonomic Keyboards have a natural shape and built-in hand-rest.

Uses of a Keyboard:

- # Keyboards are used to input data into **applications**.

For Example - Entering text into word processing applications like Microsoft Word
Entering numbers into spreadsheet applications like Microsoft Excel

- # Keyboards can also be used to **enter commands** into the computer.

For Example - Keyboard shortcuts:

- **CTRL + P** to print
- **CTRL + S** to save
- **CTRL + C** to copy
- **CTRL + V** to paste
- **PRT SC** to take a screen shot of your work



Inputting data into a computer using a keyboard.

NOTE:

Command Line operating systems require a keyboard in order to be able to use the computer at all!

Advantages/Disadvantages of using a Keyboard:

Advantages	Disadvantages
Enables fast entry of text into documents.	People with wrist and hand problems can find keyboards painful to use.
Very easy to use (you just press the keys).	Keyboards are quite large and can take up a lot of desk space .
Information that you input instantly appears on the screen . This lets you quickly check that what you are entering is correct.	Entering data is slow when compared to automatic methods. For example - a barcode code scanner will input data into the

computer almost instantly.

2 Numeric Keypads

Information:

- # Numeric keypads are used for **entering numbers into a computer system** ('Numeric' means number)
- # Some numeric keypads also allow you to enter **simple text and symbols**.

Uses of Numeric Keyboards:

- # There are many day-to-day uses of numeric keypads. Here are some of them:

ATM (Automatic teller machines) -

Entering **personal identification numbers** (PIN) into an ATM to allow us to access our cash.

Withdrawing from ATM -

Keying in **how much money** you would like to **withdraw** from an ATM.

Telephones -

These use numeric keypads to allow us to **enter phone numbers**.

Chip and Pin devices -

These have numeric keypads to allow users to **enter PIN numbers and payment amounts** when we buy goods and products.

Examples:



Bank ATM's use numeric keypads to let us enter PIN numbers and withdrawal amounts.



Mobile phones use numeric keypads to allow us to enter phone numbers.



Advantages/Disadvantages of using Numeric Keyboards:

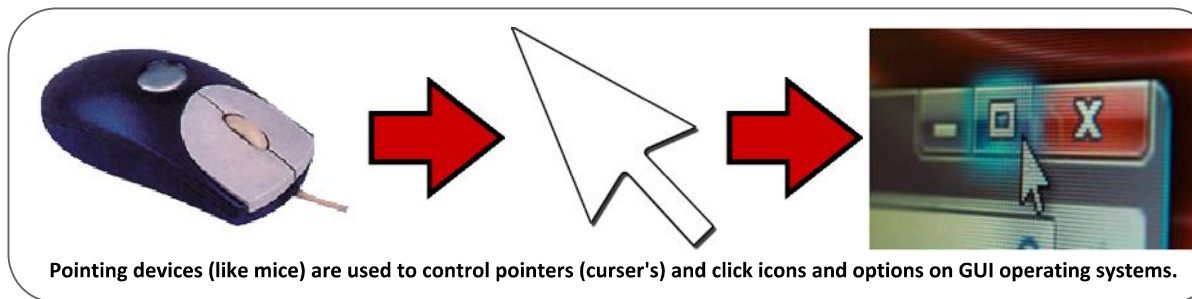
Advantages	Disadvantages
Faster than QWERTY keyboards when used for entering numbers.	People with large hands can find them hard to use because of their small keys .
Numeric keypads are small so they can easily fit on smaller devices like mobile phones.	Difficult (but not impossible) to enter text information.

Inputting data into a computer using a keyboard.

3 Pointing Devices (mouse, touchpad, trackerball)

Overview of pointing devices:

- # Pointing devices are pieces of hardware that are used to **control a pointer** (curser) on a screen.
- # Pointing devices are frequently **used with Graphical User Interface** (GUI) operating systems where pointers are used to **input commands** by selecting icons.



- # There are three main types of pointing device:

- **Mouse**
- **Touchpad**
- **Trackerball**

Information about the Mouse:

- # There are two different types of mice - a **ball mouse** and an **optical mouse**.

Ball Mouse - Uses a ball under the mouse to detect movement.

Optical Mouse - Uses reflected light to detect movement.

Examples:



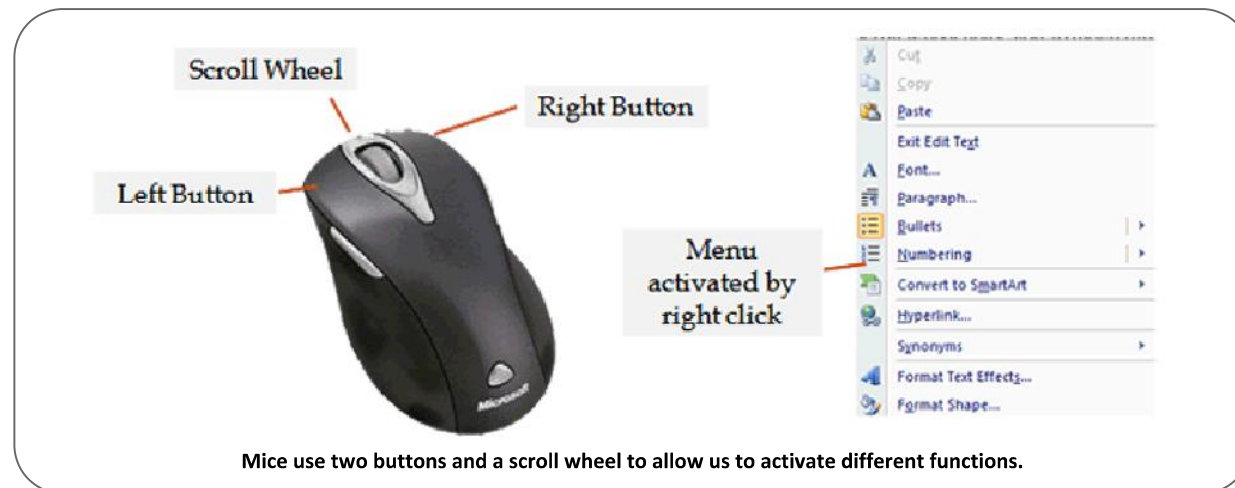
Mice are one of the devices that allow us to control a screen pointer.



Trackerballs also allow us to control a screen pointer.

Examples:

- # The ball (in a ball mouse) and the reflected light (in an optical mouse) allows the user to **direct and control an on-screen pointer**.
- # Mice have two buttons and a scroll wheel. These provide users with different functions:
 - **Left button is used to select icons and click options**
 - **Right button is used to call up a drop-down menu of options**
 - **Scroll wheel is used to move up and down through a document.**



- # **Cordless mice** work in the same way as regular mice but they **do not need a wire plugged into the computer**. They are powered by batteries instead of the computer.

Cordless mice are an example of a wireless device.

Uses of Mice:

- # Common uses of mice include:

1. **Opening, closing, maximising and minimising** programs and files.
2. **Moving, grouping and deleting** files.
3. Controlling a pointer on screen to **select icons** or **move around the page**.
4. **Editing images** in terms of size and position on the screen.



A ball mouse uses a ball to detect movement. These are prone to clogging which can affect accuracy.



An optical mouse uses light to detect movement. These are more accurate than ball mice and are better for editing images.



Cordless mice are used in exactly the same way as regular mice but they do not use a wire plugged into the computer. They use wireless technology in order to communicate with the computer. They are powered by batteries.

Advantages/Disadvantages of Mice:

Advantages	Disadvantages
Faster to select icons and options when compared to a keyboard.	People with wrist and hand problems (like RSI) can find mice painful to use.
Very easy to move a pointer around a screen and to navigate through documents.	Ball mice can pick up dirt which will affect the accuracy of pointer movement.
Mice are small and do not take up much space.	Mice need a flat surface in order to work properly For example - they would be difficult to use on train unless you had a flat table to rest on.

Information about the Touchpad:

- # Used as a **replacement to a mouse** in many laptops.
- # Users lightly **brush their finger over the touchpad** in order to control a pointer.
- # Touchpads have two buttons which have the same purpose as mouse buttons:

Left Button - Used to select icons and activate options.

Right Button - Used to call up a drop-down list of menu options.



Touchpads are built into laptops and perform the same functions as mice.

The touchpad allows us to control a screen pointer and the buttons allow us to select icons and call up options.

Uses of Touchpads:

- # Touchpads have the same uses as mice. (click [here](#) to return to mouse uses)

Advantages/Disadvantages of Touchpads:

Advantages	Disadvantages
Easier than QWERTY keyboards when used to select options and icons.	More difficult to control a mouse pointer when compared to a mouse.

Touchpads are built into laptops so there is no need for a separate mouse. This helps improve portability.	People with wrist and hand problems (like RSI) can find touchpads painful to use.
Can be used even when there is no flat surface available.	Some operations are more difficult to perform when compared to a mouse. Drag and Drop for example.
Won't get clogged with dirt (as can happen with ball mice) so performance is never reduced.	

Information about the Trackerball:

- # Trackerball's are similar to mice but the ball is on the **top of the device** instead of the bottom.
- # Screen pointers are controlled by **rotating the large ball with your hand**.
- # The buttons on trackerball's perform the same functions as those on mice and touchpads.

Uses of Trackerball's:

- # Trackerball's have the same functions as mice. (click [here](#) to return to mouse uses)
- # They are, however, **much more comfortable to use than mice**. This means that people with painful hand and wrist problems (like RSI) can use a trackerball.

Why are they more comfortable?

Trackerball's are more comfortable than mice because there is less wrist movement required. **They are more ergonomic.**

- # Some arcade gaming machines use trackerball's to control games characters on screen.
Because trackerball's are so **durable (tough)** they can deal with the wear and tear of gamers using them.

Advantages/Disadvantages of Touchpads:

Advantages	Disadvantages
People with hand/wrist problems can still use trackerball's	More expensive than mice.

Examples:

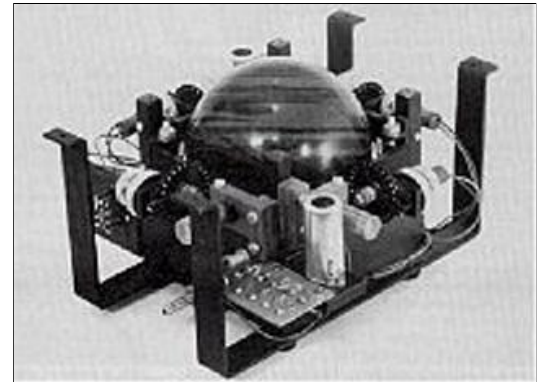


Trackerball's perform the same functions as a mouse but they require less wrist movement and so are more comfortable to use.



Old arcade games machines often used trackerball's to allow gamers to direct in-game characters.

More accurate than mice or touchpads. Pointer can be positioned more exactly which makes trackerball's very using for image editing.	More difficult to learn to use than a mouse.
Can be used even when there is no flat surface available.	
No limit on ball travel. A mouse sometimes runs out of room on a mouse mat. With a trackerball you just keep rolling.	



The world's first trackerball was developed in 1952 and used a bowling ball.

4 Remote Control

Information:

- # Remote controls are used to **control other devices** using infra-red signals.
- # Buttons on the remote control can be used to perform functions such as:
 - **Changing the channel on a T.V**
 - **Increasing/Decreasing the volume on a music player**
 - **Selecting a different chapter on a DVD player**

Uses of Remote Controls:

- # **Home entertainment systems** use remote controls. Examples are:

1. Hi-Fi systems (music centers).
2. DVD / Blu-ray Players.
3. Satellite systems.
4. Some overhead projectors (like ours in the classroom).

Examples:



A satellite T.V remote control with buttons to control the channel, volume and many more.

Industry uses remote controls to **operate machinery** that might be too dangerous to get near to. Examples include:

1. Cranes with heavy loads.
2. Activating explosives in demolition.
3. Operating robot arms in hazardous chemical plants.



A man using a remote control to operate a crane.

Advantages/Disadvantages of Remote Controls:

Advantages	Disadvantages
Devices can be operated without having to go to them . This is useful for people with disabilities.	People with limited hand movement can find them difficult to use.
They can operate devices that are in an unsafe environment . For example: Explosives to demolish a building can be set off at a safe distance.	The infra-red signal between the remote control and the device it operates can become blocked .

5 Joysticks

Information:

- # Joysticks have similar functions to that of mice and trackballs..... to control a pointer on a screen.
- # There are two main parts to a joystick:

Handle / Stick - Allows us to move a pointer in the direction we want.

Buttons - Allow us to select options or 'click stuff'.

- # Joysticks are also **popular devices for gaming**. The handle allows gamers to control on-screen characters and the buttons are used for things like firing weapons.

Examples:



A standard joystick with a control stick/handle, buttons for selecting options and a 'fire' button for gaming.

Uses of Joysticks:

- # They can control characters or objects in video games.
- # Can control **industrial machinery** (cranes for example)
- # They are used in **simulators**. For example flight simulators use joysticks in order for trainee pilots to control the simulated plane.

Advantages/Disadvantages of Joysticks:

Advantages	Disadvantages
Easier to control an on-screen curser or move an in-game character than it is when using a keyboard .	Not as easy to control an on-screen curser when compared to using a mouse .
Very simple to get used to. Anyone can just pick one up and use it!	People with hand/wrist issues (like RSI) can find them painful to use.
Can be used by disabled people. Joysticks can be operated by hands or even feet .	Difficult (but not impossible) to enter text. For example: some games allow you to select individual letters using the joystick in order to spell out your name when entering a high score.



Joysticks used to operate a crane.



Joysticks are also used to mimic real-life flight conditions on flight simulators.

6 Touch Screens

Information:

- # Touch screens allow users to **input commands into a computer** by **pressing or touching buttons/icons on the screen**.
- # Giving users this ability to touch or press icons on the screen removes the need to use a mouse.

Examples:

- # Touch screens are more suitable for use with a limited amount of options.

Uses of Touch Screens:

- # Common uses of touch screens include:

1. **Mobile phones and PDA's** use touch screens as a way of **saving space** (a separate keyboard would make the devices quite large).
2. **Public information systems** at **airports** or **tourist information offices**.
3. **Interactive whiteboards** in schools.
4. On-screen **multiple choice tests** (like driving theory tests) sometimes use touch screens.



A touch screen with 'pressable' options.



Modern mobile phones use touch screens to reduce the size of the device.

Advantages/Disadvantages of Touch Screens:

Advantages	Disadvantages
Very fast and easy to enter options (you just press them).	Limited number of options available on the screen.
No training needed in order for people to be able to use them. All people do is press the correct option.	Expensive compared to other input devices.
	Screen can become dirty and full of germs due to people touching it.



School classrooms use special touch screens called 'Interactive Whiteboards'.

7 Scanners

Information:

- # Scanners are used to **enter information on paper** (hard copies) **into a computer**.

Scanners can convert **hard copies** (printed documents, photographs etc) **into digital data**. This digital data can then be **stored and manipulated** on your computer.

- # Scanners reflect light off the hard copy in order to **capture an analogue image** of the document. The scanner then **converts the received analogue data into a digital format**.
- # Once the hard copy has been converted to digital data, it can be **stored on a computer and manipulated** using Graphics Editing Software (such as Photoshop).

Uses of Scanners:

- # Used to scan hard copy / printed documents into a computer:
- # **Old photos and important documents** (like birth certificates) can be scanned into the computer. This means you **still have a copy** if the original is damaged or lost.



Scanners are used to convert analogue hard copies of documents and photos in digital data.

Advantages/Disadvantages of Scanners:

Examples:



Earlier scanners, like the Sharp JX-450, could only scan documents in low-resolution (image quality lost).



Modern scanners are capable of scanning documents and images in much higher resolution. This means that digital copies are closer to the original quality.



Advantages	Disadvantages
Very fast to enter hard copies into the computer (compared to retyping with a keyboard).	Scanned images are usually of lesser quality than the original document.
Very easy to scan and store documents and images into the computer. Digital copies of the stored data can easily be duplicated and backed up .	Scanned images can take up a lot of disk space (especially if you scan them at a high resolution).
Old photo's that may be damaged can be scanned into the computer then repaired using Graphics Editing Software .	

Some scanners are 'hand held' which means the user physically moves the scanner across the image in order to send a copy to a computer.

8 Graphics Tablet

Information:

- # Graphics tablets are used by **artists** and **designers** to easily create and store digital graphics in a computer.
- # A graphics tablet can be drawn on with a special pen called a '**stylus**'.
- # Whatever is drawn onto the tablet can be seen on a computer screen.

It is really just the same as drawing on paper but the final image is **digital**.

- # Anything drawn onto the tablet can be saved into the computer as images.

Once images are saved into the computer, they can be **edited using graphics editing software**.

Uses of Graphics Tablets:

- # Allow designers to produce digital images **much more accurately** than if they were using a mouse.

Examples:



A graphics tablet with a stylus pen .



Graphics tablets can be used to accurately 'draw' illustrations, symbols and diagrams.

Advantages/Disadvantages of Graphic Tablets:

Advantages	Disadvantages
Much easier to draw images using a pen than it is to draw them using a mouse.	Very expensive when compared to a mouse.
Very accurate drawings can be produced. Stylus pens make it better for the artist to capture fine detail than they could using a mouse.	Not useful for some input needs. For example - graphics tablets are not designed for entering data into the computer by clicking icons or menus.
Useful in countries where drawings and symbols are used instead of text (like China and Japan).	



A high-end illustration of a car produced using a graphic tablet.

9 Light Pens

Information:

- # Light pens are quite old technology and can be used as **alternatives** to **mice**, **touch screens** or **graphics tablets**.
- # Light pens are used by **directly drawing or selecting icons** on a **computer screen**.

Note: Currently light pens can currently **only be used with CRT** (cathode ray tube) monitors.

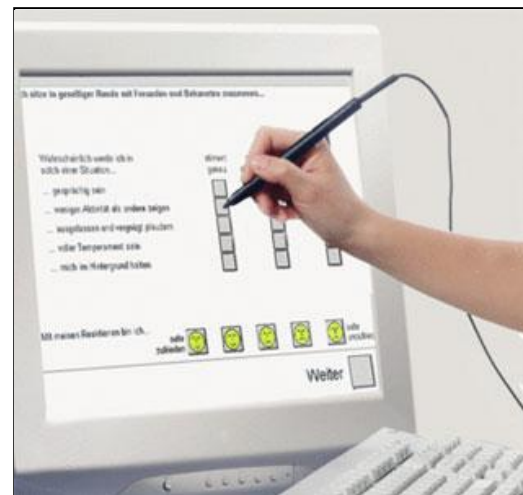
- # They work by **picking up light from the computer screen** and then signaling to the computer exactly where the light was picked up. This allows the computer to '**draw**' on the part of the screen where the pen picked up the light.

Uses of Light Pens:

- # Used for **selecting objects** on a **CRT screen**.

For example: replacing a mouse or touch screen clicking/touching icons.

Examples:



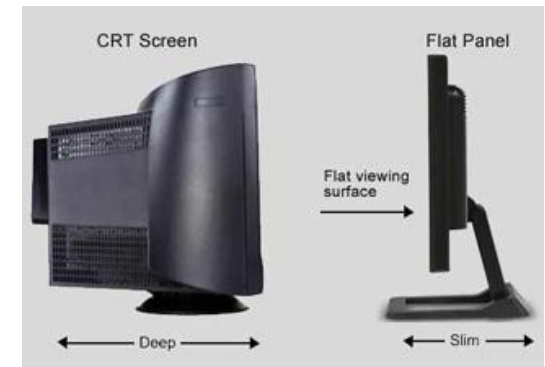
Light pens can replace mice and touch screen for selecting items on a CRT screen.

Used for **directly drawing** onto a **CRT screen**.

For example: replacing a graphics tablet to produce hand-drawn designs.



Light pens can be used to produce drawings directly on a CRT screen.



Light pens only currently work with CRT Screens (the fat ones) .

Advantages/Disadvantages of Light Pens:

Advantages	Disadvantages
More accurate than touch screens. (end of the pen is much thinner than a finger and so can accurately select small icons)	Not as accurate as a graphics tablet when used for producing drawings.
They are very small and so can be used where there is no room for a mouse or graphics tablet.	Can only be used with CRT (cathode ray tube) screens. (not usable with TFT or LCD screens)
	Can be quite uncomfortable to use as you have to hold your hand raised up to the computer screen for long periods.

10 Microphones

Information:

Microphones are used to **input analogue sounds** into a digital computer.

1. Sounds exist in the form of **sound waves** (analogue signals that change slowly).
2. Computers understand **digital data** and **not analogue**.

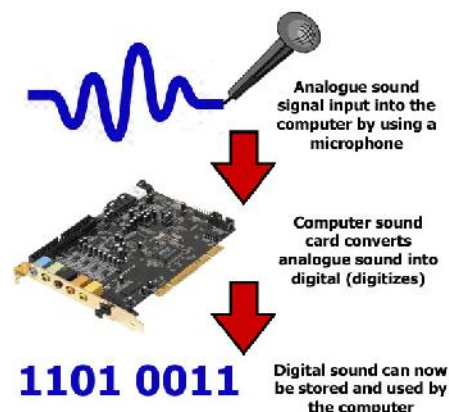
Examples:

For example: computers understand combinations of **0's** and **1's**.

3. The job of the microphone is to input analogue sound signals into the computer where the sound card will convert analogue sound into digital (in other words the sound card will **'digitize'** the analogue sound into 1's and 0's)

Note: the sound card acts as an **ADC** (analogue to digital converter).

4. Once digitized, the sounds can be stored and used by the computer.



Microphones are used to input sound into a computer.

Uses of Microphones:

Used to **input sounds/speech** for use in a range of applications. For example:

- **Narration** (spoken words) **in presentations or in web sites**
- **Voice-over's in movies**
- **Speaking over the Internet using VoIP** (Voice over Internet Protocol)
An example of this would be using Skype to hold an online conversation
- **Conducting videoconferencing** (e.g. business meetings online).

Used in **voice recognition software**. For example:

- **Converting speech into text** for use in word processors (useful for people who do not have use of their hands)
- **Issuing commands** (e.g. some cars allow for speech to activate certain features like **'starting the engine'** or **'altering radio volume'**).

Hands-Free mobile phones use a microphone to allow people to hold conversations without using their hands (this makes driving much safer).

Used (along with headphones) in **gaming** so that gamers can talk to each other.



Microphones can be used to hold voice conversations over the Internet (like using Skype).



Microphones are used on hands-free mobile phone kits to make driving safer.

Advantages/Disadvantages of Microphones:

Advantages	Disadvantages

Faster to read in text than to type it in using a keyboard .	Sound files can take up a lot of computer storage space.
Means that word processors can be used by people who cannot use their hands. Words are simply spoken into the software instead of typed .	Speaking words into a word processor is not as accurate as typing . This means that mistakes are common with words that sound the same but have different meanings. For example: ' see ' and ' sea '.
Makes driving safer because people do not need to take their hands off the steering wheel in order to use their mobile phone.	Background noise can interfere with voice recognition systems.
Microphones are very cheap to buy.	



Gamers use microphones to communicate instructions to each other in-game .

11 Digital Cameras

Information:

- # Have pretty much replaced the old style film-based cameras.
- # Digital cameras **store digital photographs on a memory card** and these images can easily be transferred onto a computer for viewing or editing.
- # Most digital cameras have a **built-in screen to allow for previewing** the digital photograph. This is useful so that the photographer can decide which images to keep and which to delete.
- # Digital camera image quality is measured in '**Megapixels**'. These are the **number of dots** that make up the image. Generally speaking, the **higher the Megapixels**, the **higher the detail quality** of the image.

3 Megapixel cameras produce images made up of **3 Million pixels** (dots).

24 Megapixel cameras produce images made up of **24 Million dots** (8 times more detail)

- # Most digital cameras can also **capture video and sound** as well as **images**.

Examples:



An old film-based camera .

- # Almost all **modern mobile phones** have a built-in digital camera.

Uses of Digital Cameras:

- # Used to **capture digital images** for uses such as:
- **Saving special moments** (weddings, birthdays, family snaps etc)
 - **Gathering images for use in presentations, web pages or publications** (newspapers and magazines etc)
- # Easy to **transfer images** to a computer for **editing**.
- For example:** using PhotoShop to alter brightness or contrast
- # Images can be **printed easily** by:
- **Uploading to a computer and then printing**
 - **Connecting directly to a photo-printer and printing straight from the camera** (no need to upload to a computer)
- # Most digital cameras also allow for short, **high-quality video clips** to be produced.



A modern digital camera with preview screen.



Traditional photographs were stored on reels of film .

Advantages/Disadvantages of Digital Cameras:

Advantages	Disadvantages
No film to develop. This means that producing the photo is much quicker.	Camera user needs to have computer skills to be able to transfer and print digital photographs.
Unwanted images can easily be deleted from the memory card.	Not all digital cameras produce images as high-quality as traditional cameras.
Because the images are digital it is easy to make copies of them.	Artistry of traditional photography is being lost .
No need to print at all because the images can be viewed a computer screen. This saves paper and is better for the environment .	Camera software can automatically correct 'bad images' making photography easier to master without as much skill .



Digital photographs are stored on memory cards.

Memory cards can **store thousands of digital photographs**. Traditional photographs can take up a lot of physical storage space.

Digital images can **easily be transferred** using **bluetooth, email attachments** and **mobile phones**.

Digital images can be **improved and edited easily** using software (sharpening the image for example)

People using traditional 'film-cameras' had to take much more care to capture each image perfectly.



Traditional photography was much more difficult to master. Much thought had to go into capturing quality photographs.

Digital photographs are run-through software which does most of the hard work for us and makes it harder to produce bad images.

12 Web Cameras (web cams)

Information:

- # Web cameras are **similar to digital cameras** in their function but they are **directly connected to a computer** and **do not have memory storage**.
- # Web cameras can capture both **digital images** and **video**.
- # Images / video are **sent directly to the computer** where they can be stored and used.

These digital images and video are often added to **presentations, web sites** and other publications.

YouTube channels sometimes use web cams to **capture the content of the channel**. This is usually a presenter giving out information about whatever topic the channel is about.

- # Many modern computing devices (such as laptops) come with **built-in web cams**.

Examples:



A web cam attached to a computer monitor.

Uses of Web Cameras:

- # Used to conduct **face-to-face conversations** with friends and family online over **VoIP** applications such as **Skype**.

A similar use is to conduct **face-to-face job interviews over the Internet**. This removes the need for job applicants to travel very large distances to attend the interview.

- # Used to hold **video-conferencing** meetings over the Internet.
- # Can **capture image and video content** for presentations, web sites and even YouTube content.
- # Web Cameras are sometimes used by **parents to keep an eye on their children** when they are in another room.
- # Some people use web cams as a cheap **alternative to security cameras**.
- # Sometimes used to allow drivers to **view the traffic conditions on roads**.

This allows the driver to plan a different route if the traffic on a road is too congested.



Web cams allow long distance face-to-face interviews to take place.



Web cams allow elderly people to hold face-to-face conversations with relatives.

Advantages/Disadvantages of Web Cameras:

Advantages	Disadvantages
Allow for long-distance face-to-face conversations . This removes the need for people to travel. (especially useful for elderly and disabled people to keep in touch with family)	The image/video quality can sometimes be poor (low resolution).
Video conference business meetings and job interviews can be conducted very cheaply over long distances.	They do not have their own storage and so must be permanently fixed to the computing device.

(no need for expensive travel or bookings of hotel rooms to attend the meeting or interview)

Parents can keep an eye on their children from a different location.

Web cams can be **left running constantly** which makes them useful as **security devices on properties**.

(as a form of security camera)

Are usually in a **fixed position** and so cannot be easily moved around.



Traffic cams allow drivers to avoid heavy traffic.

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